Exercise 1

LLMs for the Economic and Social Sciences

Today

Overview

Coding setup:
- Python
- Jupyter notebooks
- IDEs

Collaboration:
- Version control
- GitHub
- Package management

Computing resources:
- BWUniCluster 3.0
- Google Colab

Introductions

Please briefly introduce yourselves

- Name
- Course of study
- Why you are interested in the course

Previous experience

https://ilias.uni-mannheim.de/vote/JDVI

Basics

Python

We will use Python throughout this course, and you will also need to use it for your projects.

https://www.python.org/

If you are not yet comfortable with Python, then you must invest time getting up to speed as soon as possible. An example tutorial can be found in the documentation.

Basics

Coding

When writing code, we often want more than just a text editor. There are many options with different features.

Jupyter notebooks enable interactive coding.
https://jupyter-notebook.readthedocs.io/en/latest/

There are also different computer programs that you can use called IDEs. These can help with highlighting errors in your code or making suggestions for code completion. Feel free to explore the options.

Basics

Coding best practice

1. Write documentation in the ReadMe for how someone would use your code
2. Include comments in the code
3. Use functions if you need to reuse the same code chunks
4. etc.

You can find more detailed suggestions online e.g. from ruimaranhao

Version control

What is it? Why do we use it?

Version control enables the reversal of unwanted code changes and keeps track of progress.

It can be useful when coding alone, but it especially important when collaborating. Using version control means that multiple people can work on the code at the same time and then handle merging all the changes together.

A record of each version is maintained, meaning that you can also revert back to previous versions if needed.

Version control

GitHub

One of the most popular methods of implementing version control is using GitHub.

https://docs.github.com/en/get-started/start-your-journey/about-github-and-git

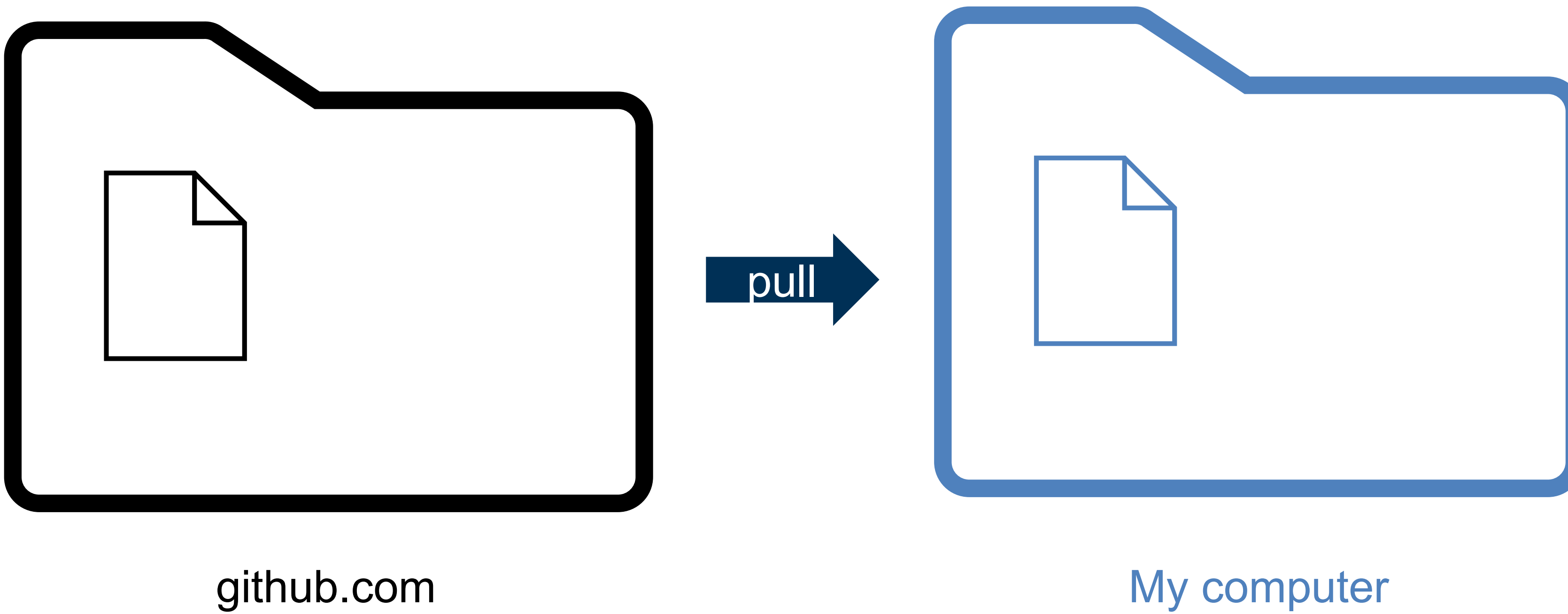
1. Create an account
2. Set up Git on your computer

Each code project is stored in a repository. There is a *remote* version of this online, but each user can also store a *local* version on their computer.

Now we will look at how to use GitHub theoretically, and then at a practical example.

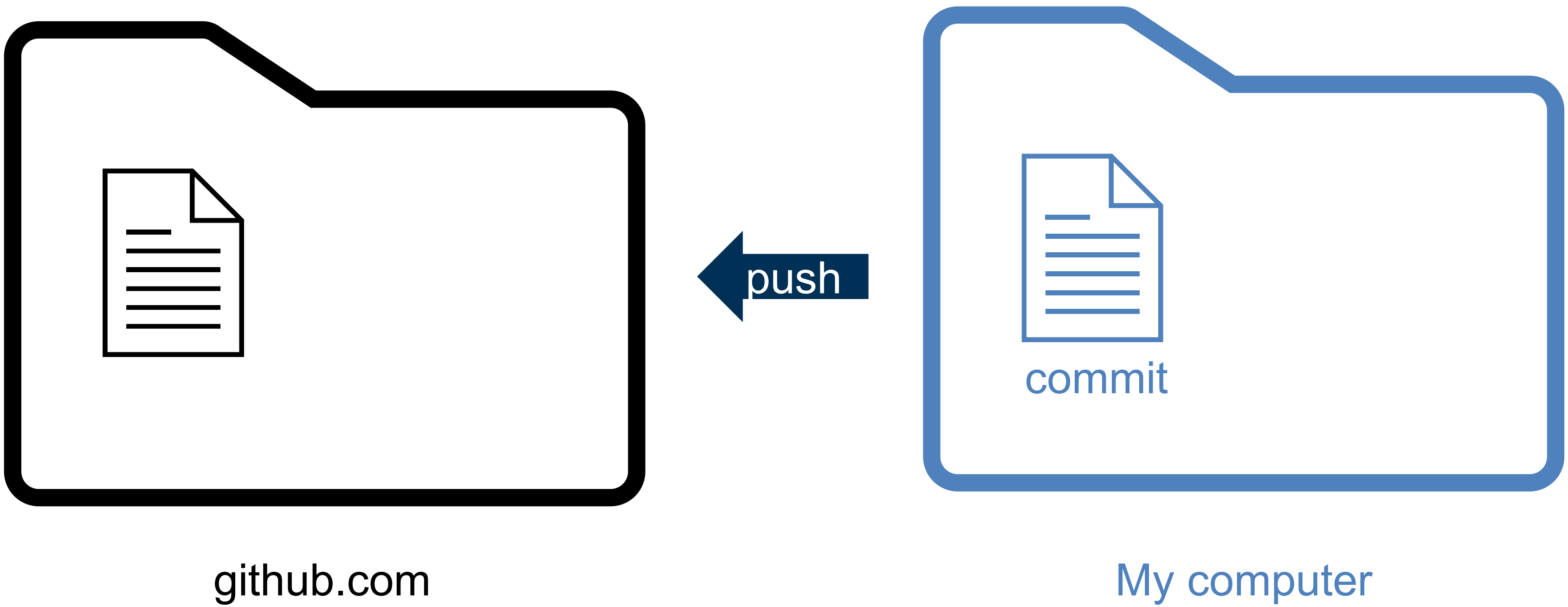
GitHub

Repository



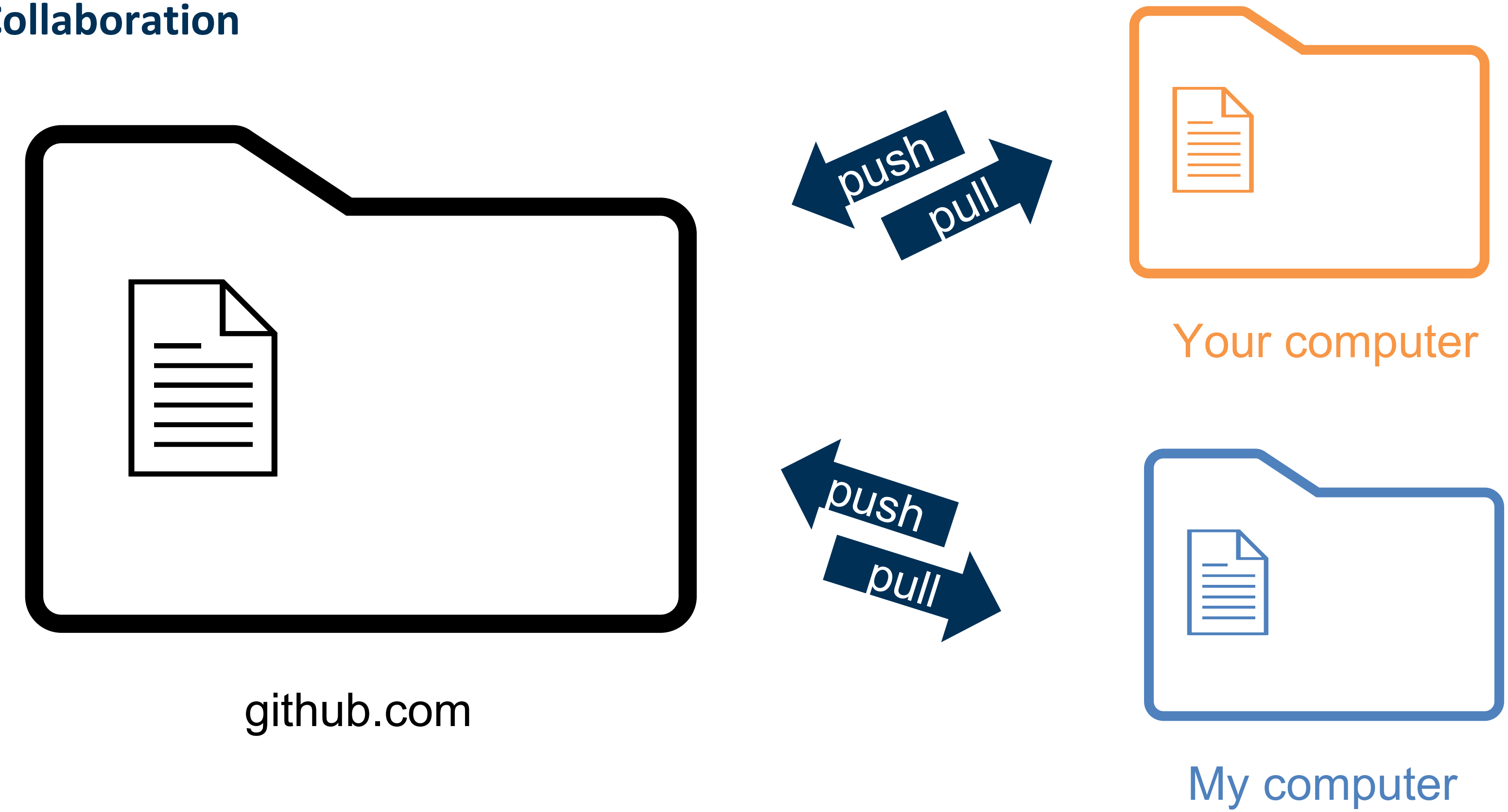
GitHub

Making changes



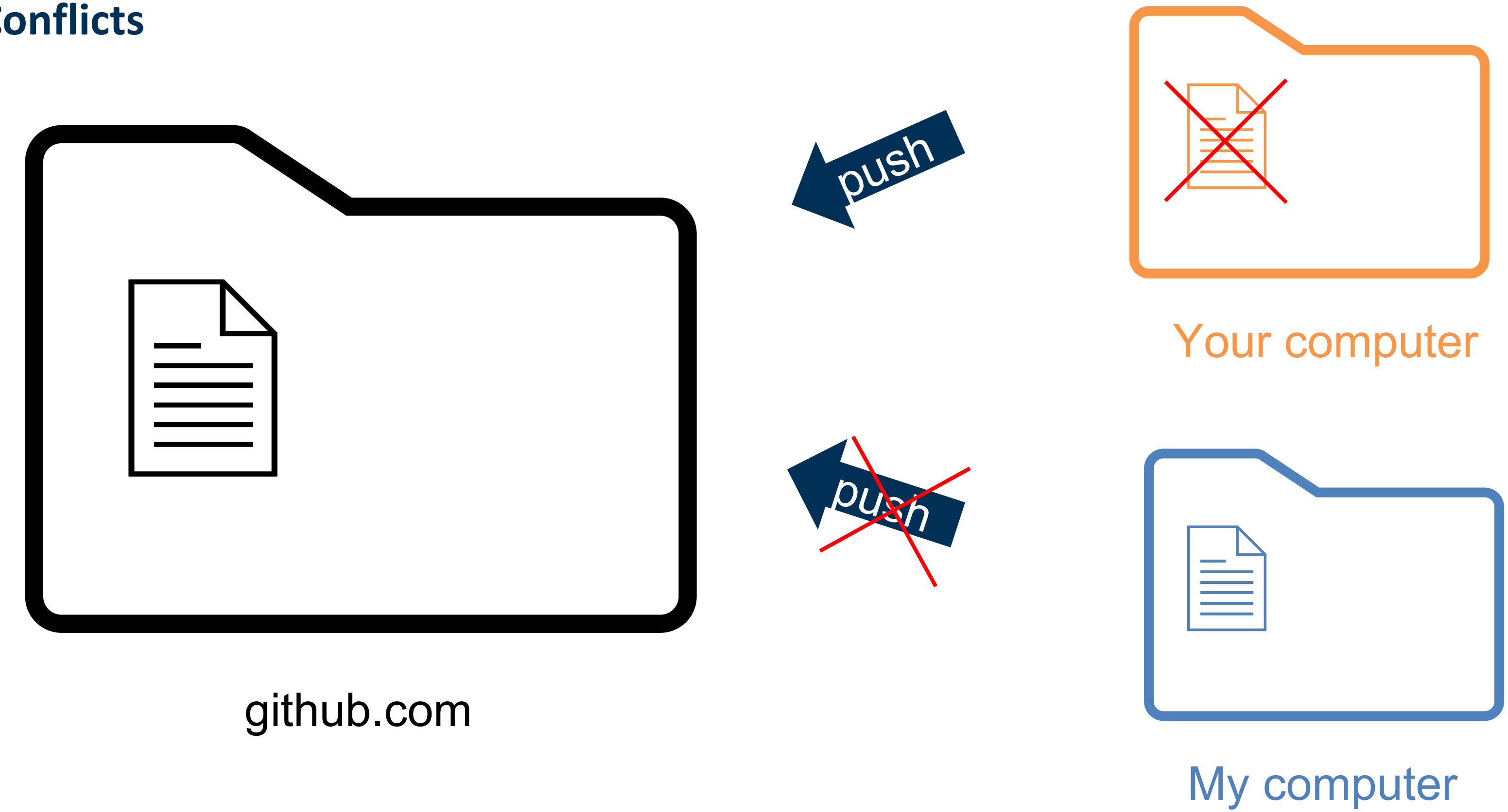
GitHub

Collaboration



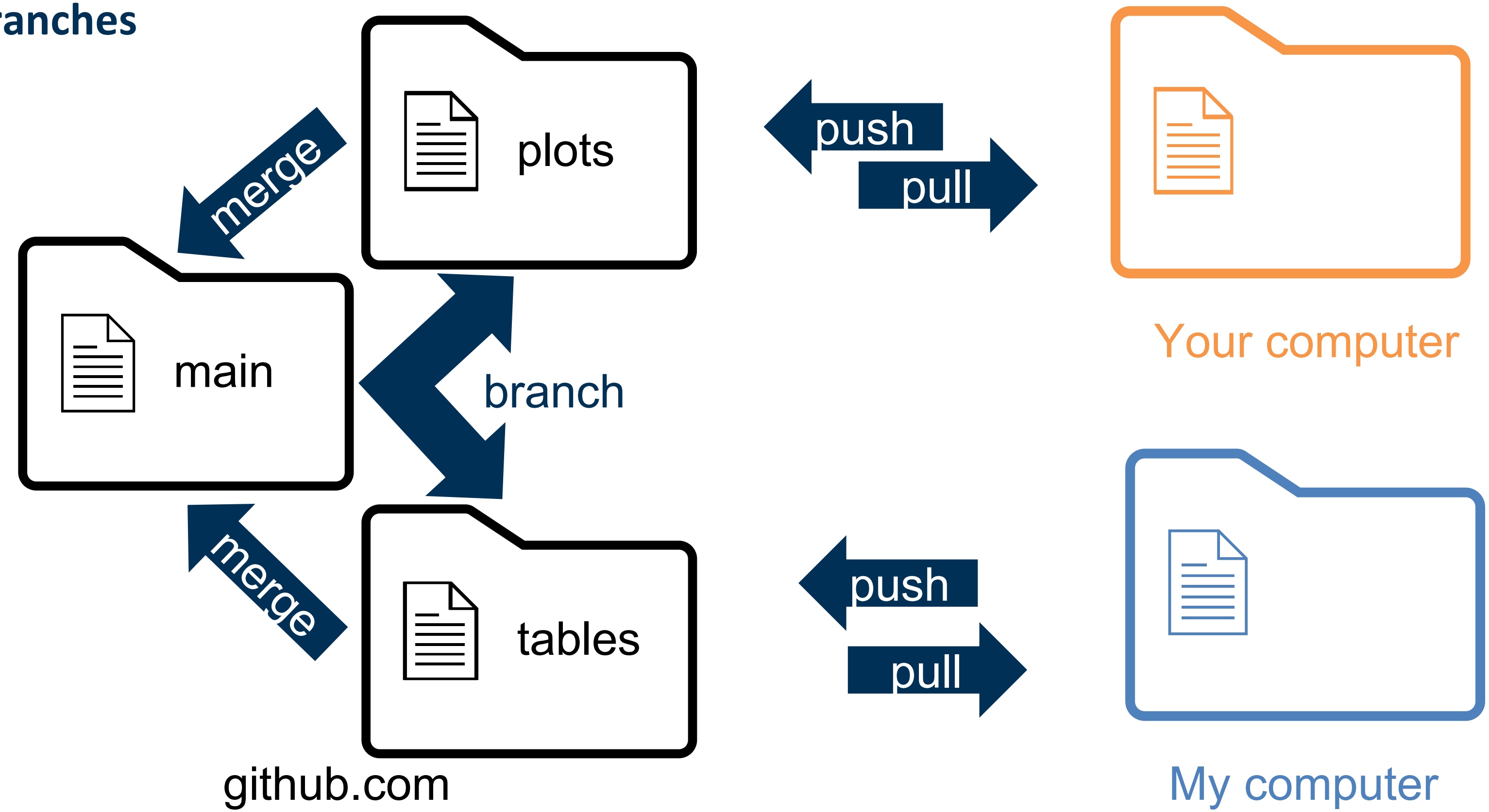
GitHub

Conflicts



GitHub

Branches



GitHub

Have a go!

As a group, you will be allocated a file, where you should all add a similar analogy about LLMs

1. Clone the practice repository to your computer
2. Create your own branch
3. Add your own sentence to the 'practiceX.md' file and commit the new version
4. Push the latest version of the branch
5. Create a pull request for merging your branch into the main branch
6. Within your group
    1. Review someone else's pull request
    2. Check that only the 'practiceX.md' file is modified
    3. Approve and merge the pull request if you are happy. You will need to handle merge conflicts

See https://learn.github.com/skills for more guidance

Basics

Package management

Beyond basic Python, we will use additional packages e.g. NumPy.

When collaborating as a team, you should make sure you are using the same versions of packages, as functionality can change between versions. Options for handling this include:

conda: Anaconda and Conda Forge

venv

uv

Computing resources

Why do we need additional resources?

LLMs range in size but generally require more computing resources than a standard laptop.

They can be accessed via APIs if hosted elsewhere or run on GPUs.

Additional computing resources could be accessing models via an API or hosting them yourself on GPUs.

Computing resources

BWUniCluster 3.0

https://wiki.bwhpc.de/e/BwUniCluster3.0/Getting_Started

High Performance Computing resource available through your membership of the university. After registration, you can request time using GPUs to run your experiments.

Getting access:

A. Request entitlement via the university - Request entitlement
B. Within 14 days of receiving entitlement, complete the questionnaire – Questionnaire
C. Register for an account on the cluster - Registration

Can be used via Slurm or a Jupyter notebook

Computing resources

Google Colab

https://colab.google/

Uses Jupyter Notebooks and enables access to the Gemini API and to GPUs

Computing resources

GitHub Education

https://github.com/education/students

You can get access to resources if you link your student email address and request access
Including free access to a coding agent, which will be relevant in a few weeks

Practical

Text processing in Python

Now to 1_python_basics.ipynb…